Language-Guided Aerial Navigation with Conditional Task Graphs and Joint Task–Viewpoint Planning

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***Abstract*—Long-horizon aerial inspection requires efficient travel among multiple targets while respecting task dependencies, acquiring visual evidence, and responding to missing objects. Existing vision-and-language navigation methods often exhibit low success rates on long-horizon, multi-subtask missions, perform redundant actions that reduce execution efficiency, and have limited ability to handle unexpected situations. We present LogicFly, a language-guided aerial navigation framework that combines a reusable hierarchical semantic–geometric map with conditional task reasoning and observation-aware planning. A large language model grounds instructions in the map to construct a Conditional Partial-Order Task Graph (CPTG) that separates mandatory precedence from observation-triggered branches and distinguishes navigation anchors from verification targets. Joint Task–Viewpoint Planning (JTVP) jointly optimizes task order and three-dimensional observation poses within a fixed-size receding horizon. By optimizing a bounded task window and executing only the first visit before replanning, LogicFly limits the combinatorial planning problem while supporting many subtasks within one mission. Online verification and language-model-guided semantic-region recovery update task states after negative observations, while safe-corridor-constrained trajectory optimization supports aerial execution. Across 854 simulation episodes in six indoor scenes, LogicFly achieves a 96.96% strict full-task success rate and 0.8394 success weighted by path length. An additional simulation completes all 15 subtasks of a single instruction within the evaluation time budget. Physical UAV experiments further demonstrate multi-target execution with conditional logic and recovery in indoor environments.**

***Note to Practitioners*—LogicFly is applicable to search tasks in a wide range of indoor environments whose structural layouts remain relatively stable. An operator may issue a long natural-language request containing several checks, mandatory precedence relations, freely ordered subtasks, and observation-dependent alternatives. The framework converts such requests into a conditional partial-order task graph and jointly selects the order and three-dimensional camera viewpoints of currently executable tasks. Negative visual evidence can trigger bounded semantic-region search and replanning rather than immediate failure. Its practical benefit is to reduce unnecessary travel while preserving the logic of the request and acquiring task-specific visual evidence. Deployment requires a previously constructed semantic–geometric map, calibrated RGB-D sensing, and localization in the map frame. The demonstrated platform uses an NVIDIA Jetson Orin NX (16 GB), a PX4 flight controller, a Livox Mid-360 LiDAR, and an Intel RealSense D435 depth camera, with FAST-LIO localization. Performance depends on map quality, perception reliability, and the coverage of candidate viewpoints; changes that invalidate the stored map may require renewed exploration or online recovery.**

***Index Terms*—Aerial robotics, task and motion planning, vision-and-language navigation, semantic mapping.**

I. Introduction

As unmanned aerial vehicles (UAVs) are increasingly deployed in intelligent building inspection, indoor security, and emergency response, enabling autonomous navigation guided by complex natural-language instructions has become an important research challenge. Real-world missions often involve multiple spatial regions and target objects, subject to ordering constraints, conditional dependencies, and dynamic task requirements. Successfully executing such missions requires UAVs to go beyond conventional perception and motion planning by integrating spatial-semantic representations, efficient multi-goal task planning, and logical reasoning. Effective task planning is also essential to minimize redundant travel and maximize mission efficiency within limited flight time. Moreover, visual occlusions, uncertain target locations, and environmental changes demand adaptive decision-making based on real-time observations, rather than reliance on predefined task sequences. Vision-and-Language Navigation (VLN) benchmarks such as R2R [1] and VLN-CE [2] have advanced instruction following in discrete and continuous environments, while ALFRED [3] and long-horizon VLN [4] extend embodied reasoning toward compositional tasks. Nevertheless, most existing formulations remain centered on a single goal, a prescribed route, or a short sequence of actions. They do not directly address long-range aerial instructions that combine multiple subtasks, logical dependencies, spatial relations, and observation-dependent alternatives.

Recent language-model-based VLN methods use observation histories [5], map-guided prompting [6], or structured spatial memory [7] for navigation. Persistent semantic representations provide reusable spatial memory and target anchors [8]–[10]. Despite these advances, existing approaches still struggle with complex logical constraints, multi-target coordination, and redundant exploration in large-scale environments. Their predominantly goal-by-goal decisions leave task dependencies and observation-driven changes insufficiently integrated with geometric planning. Long-horizon aerial missions therefore require coordinated task ordering, informative observation viewpoints, and recovery from missing or occluded targets to improve both full-task success and execution efficiency.

To address these challenges, we propose **LogicFly**, a language-guided aerial navigation framework for long-range, multi-subtask instructions with complex logical dependencies, as illustrated in Fig. 2. LogicFly comprises four collaborative modules. First, autonomous exploration constructs a

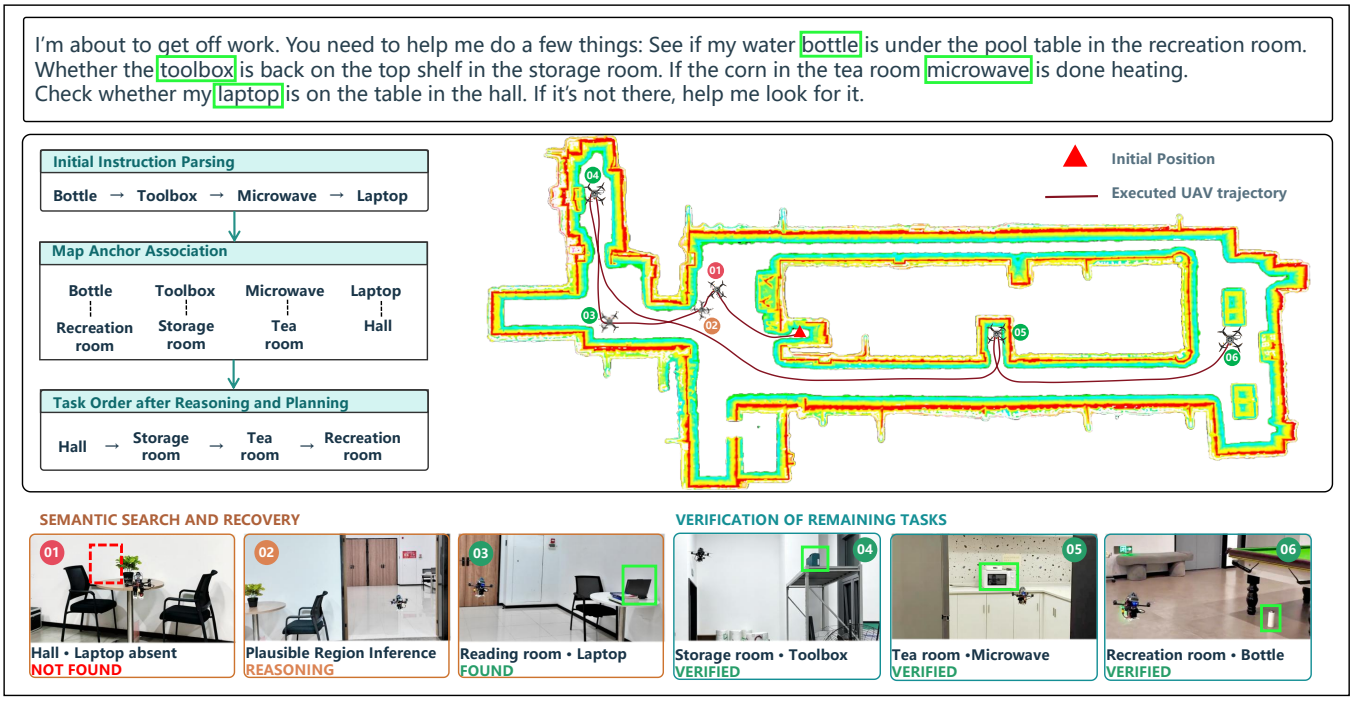

Fig. 1. Real-world execution of a multi-target instruction by LogicFly. The UAV first fails to detect the laptop at its instructed location, performs semantic recovery, and then verifies the laptop, toolbox, microwave state, and water bottle. The map panel shows the executed route, while numbered views document the recovery and task-verification sequence.

reusable hierarchical 3D semantic–geometric representation of free space, floors, rooms, cross-floor transitions, and open-vocabulary object instances. Second, a large language model grounds the instruction in the semantic map to construct a **Conditional Partial-Order Task Graph (CPTG)** of executable subtasks while separating mandatory precedence from observation-triggered branches and navigation anchors from verification targets. Third, **Joint Task–Viewpoint Planning (JTVP)** jointly selects executable-task order and collision-free 3D observation poses within a receding horizon, accounting for travel cost, visibility, and terminal observation quality. Fourth, a closed-loop perception and execution module verifies task outcomes, generates smooth trajectories, and invokes bounded recovery after negative observations through local viewpoint retry, language-model-guided semantic-region search, belief revision, and replanning. Online evidence revises downstream decisions while preserving task dependencies. After execution, spatially consistent multi-view detections incrementally enrich the semantic map with newly discovered object instances for subsequent missions.

We evaluate LogicFly on 854 valid episodes generated from 152 instruction cases across six HM3D scenes. The benchmark includes single- and multi-target tasks, single- and cross-floor navigation, precedence constraints, conditional branches, spatial relations, verified absence, semantic recovery, and target disambiguation. LogicFly achieves a 97.0% strict full-task success rate and an SPL of 0.84, including 96.70% success on multi-target tasks. Ablation studies show that CPTG is critical to reliable multi-target execution, while JTVP increases SPL by 5.33 percentage points and reduces mean successful-task completion time by approximately 6.6% at the same success rate. We further deploy LogicFly on a physical UAV in multiple real indoor environments, demonstrating reliable execution of long-range, multi-subtask navigation instructions with complex logical dependencies.

The main contributions of this paper are summarized as follows:

- We present **LogicFly**, an end-to-end aerial navigation framework that reuses an autonomously constructed hierarchical semantic–geometric representation to execute long-range instructions containing multiple subtasks and complex logical dependencies.
- We propose a **Conditional Partial-Order Task Graph (CPTG)** that explicitly represents mandatory precedence, freely schedulable tasks, observation-triggered branches, and distinct navigation and verification targets.
- We develop **Joint Task–Viewpoint Planning (JTVP)** and an observation-driven recovery mechanism to couple task ordering with 3D evidence acquisition, reduce geometric travel, and respond autonomously to negative observations.
- We conduct extensive simulation and real-world evaluations, demonstrating high strict full-task success, improved efficiency over representative baselines, and reliable deployment on a physical UAV in complex indoor tasks.

II. Related Work

A. Long-Horizon and Multi-Target Vision–Language Navigation

Early VLN benchmarks established instruction following along a route [1], [11], object grounding [12], [13], and continuous-space navigation [2]. PREVALENT [14] and ScaleVLN [15] improve learned navigation through multimodal pre-training and large-scale instruction–trajectory generation, while HAMT [16] and DUET [17] preserve temporal or graph-structured navigation history. These methods substantially improve route following, but their task abstraction remains dominated by a single destination or a sequential instruction whose action order is largely fixed.

Subsequent work extends navigation toward longer and repeated missions. MultiON [18] evaluates semantic-map memory through a prescribed sequence of object goals and shows that performance falls sharply as task complexity increases. M-TRACK [19] decomposes long ALFRED instructions into sequential milestones and checks progress before advancing, whereas LH-VLN [4] introduces multi-stage tasks and dynamic memory for long-horizon consistency. Iterative VLN [20] studies experience reuse across episodes, and GOAT-Bench [21] evaluates lifelong navigation to sequences of open-vocabulary goals specified by multiple modalities. OneMap [22] further retains open-vocabulary semantic observations across consecutive object searches. These advances establish the importance of persistent memory and multi-goal evaluation; however, their subgoals are generally presented in a prescribed order or handled as consecutive independent queries. Their sequential goal interfaces provide limited support for scheduling independent subtasks and revising downstream task logic from verification outcomes. LogicFly addresses these limitations with CPTG, which preserves mandatory dependencies while retaining freedom among unordered subtasks and activating or skipping branches from online observations. JTVP optimizes task order and observation poses within a fixed-size window, executes the first visit, and replans from updated evidence, bounding the combinatorial optimization by the planning horizon.

B. Semantic–Geometric Memory and Zero-Shot Navigation

Persistent maps reduce repeated exploration by separating environmental knowledge from the current query. 3D Dynamic Scene Graphs [23] and Hydra [24] organize metric geometry, places, rooms, objects, and agents in hierarchical representations, while OpenScene [25] supports open-vocabulary queries over 3D features. VLMaps [8] grounds spatial language in language-aligned maps; ConceptGraphs [9] and HOV-SG [10] extend this idea to open-vocabulary object graphs and building-scale hierarchies. Collectively, these representations provide reusable target anchors and spatial context, but a map alone does not specify how several logically related observation goals should be scheduled or verified.

LLM- and VLM-based navigation agents exploit such context without task-specific VLN training. LM-Nav [26] composes pretrained language, vision, and navigation models, and L3MVN [27] uses language-model commonsense to select semantic frontiers. NavGPT [5] reasons over observations and navigation history, while MapGPT [6], SpatialGPT [7], and SpatialNav [28] use explicit maps or scene graphs to improve spatial reasoning. VLN-Zero [29] combines rapid exploration with compact scene-graph memory, and OpenNav [30] brings zero-shot reasoning to continuous environments. Most recently, MSGNav [31] preserves visual evidence in a multimodal 3D scene graph and introduces visibility-based terminal-viewpoint selection. This progression improves spatial memory, exploration, and final-goal visibility, yet typically optimizes the next action or viewpoint for one active goal. Efficient multi-target execution remains a challenge: goal-by-goal planning underuses ordering freedom, while target-not-found observations require coordinated changes to task logic and search. LogicFly addresses these limitations with JTVP and CPTG. JTVP jointly selects executable tasks and 3D observation poses using collision-aware travel costs. CPTG retains resolved outcomes, pending tasks, prerequisites, and conditional states as compact structured memory across planning cycles. Negative evidence updates regional confidence and triggers local or semantic-region recovery, reducing redundant movement and improving robustness to missing targets and stale semantic anchors.

C. Language-Grounded Task and Motion Planning

Scene-graph planning connects high-level language semantics with geometric feasibility. SayPlan [32] searches hierarchical 3D scene graphs, combines LLM planning with a classical planner, and uses simulator feedback to revise infeasible plans. OSGP [33] translates natural-language missions into linear temporal logic and performs optimal hierarchical search with LLM guidance. Ray *et al.* [34] derive sparse task-and-motion planning domains from hierarchical 3D scene graphs, reducing computation on irrelevant objects. Complementarily, Inner Monologue [35] shows that success detection and environment feedback improve closed-loop language-model planning.

These approaches address increasingly expressive task specifications and provide feedback at the symbolic or motion-feasibility level. However, they generally treat semantic objects or symbolic actions as planning goals rather than task-conditioned visual observations. Consequently, camera-pose quality, collision-aware travel between alternative observation poses, and recovery from target-not-found evidence are not optimized together. LogicFly closes this gap through a geometry-aware execution abstraction: CPTG preserves precedence and conditional logic, JTVP jointly selects executable tasks and 3D observation poses, and online evidence updates both task state and semantic-region search. Coupled with collision-free trajectory generation, this design turns language-level plans into efficient closed-loop aerial execution and prevents a single negative observation from prematurely terminating the overall instruction.

III. System Overview

Figure 2 summarizes the information flow in LogicFly. The system takes a natural-language instruction, a reusable semantic–geometric map, and the current UAV state, and

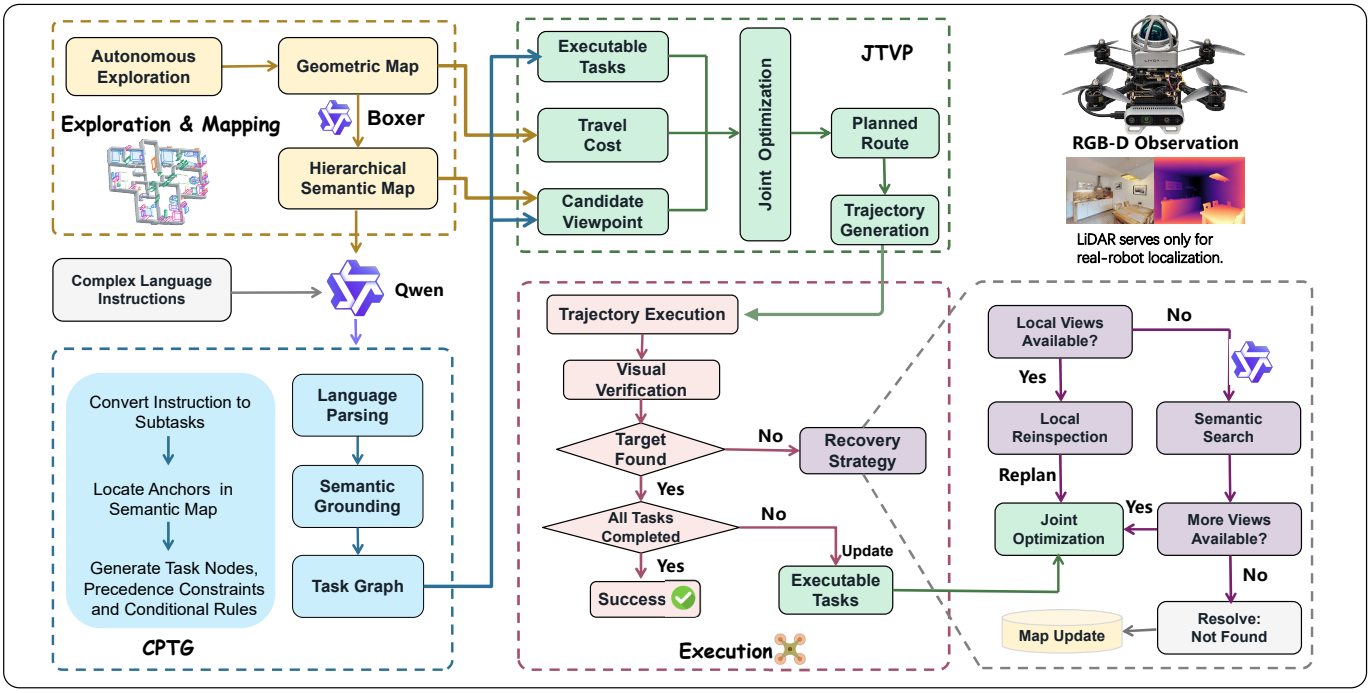

Fig. 2. Overview of LogicFly. A natural-language instruction and an autonomously explored hierarchical semantic–geometric representation are converted into a CPTG and an executable task set. JTVP jointly selects task order and 3D observation poses within a receding horizon. The first planned visit is then executed and verified online; the resulting observations update the task and search states, revise semantic map priors, and trigger recovery and replanning when necessary.

returns verified task outcomes through repeated planning, flight, and observation.

The reusable 3D semantic–geometric map supports task reasoning, planning, and recovery. Autonomous exploration provides registered RGB-D observations and occupancy geometry. Fused 3D objects are organized into floors, rooms, and functional regions, with cross-floor connectivity linking navigable areas. Semantic anchors ground language references, observation regions, and alternative search locations; occupancy and free space support collision-aware travel costs and trajectory generation. Online verification revises regional confidence and coverage for replanning. Spatially consistent multi-view detections are fused into new object instances after execution, enriching the map for subsequent missions.

A large language model takes the instruction and a compact inventory of mapped rooms and objects as input and produces a structured CPTG that separates precedence constraints from conditional activation. Each task specifies where to navigate and what evidence to verify; deterministic checks validate entity references and task dependencies before execution. From the currently executable tasks, JTVP jointly selects task order and 3D observation poses within a bounded window. Only the first visit is committed before the task and map states are reconsidered.

During execution, SAM 3.1 [36] and Boxer provide target detections and 3D localization, while safe-corridor-constrained MINCO optimization converts the selected route into a smooth UAV trajectory. Terminal verification updates CPTG outcomes. A negative observation can trigger local reinspection or search in alternative semantic regions; the resulting coverage and confidence changes feed back into the next planning cycle.

IV. METHODOLOGY

This section defines the map and task representations, derives the observation and motion costs used by JTVP, and specifies trajectory optimization and observation-driven recovery.

A. Reusable 3D Semantic–Geometric Map

The robot first explores an initially unknown environment and constructs a reusable hierarchical semantic–geometric representation from onboard observations. The geometric component $\mathcal{M}_g$ records occupied, observed-free, and unknown 3D space, while the semantic component $\mathcal{M}_s$ organizes floors, rooms, cross-floor transitions, and open-vocabulary object instances in a shared global frame. We use FALCON [37] for autonomous aerial exploration; the resulting trajectories support the acquisition of RGB-D observations, camera poses, and occupancy evidence. The updated perception and room modules below convert these observations into auditable object instances and supported room regions without filling unknown space as free or silently assigning it to a room. This representation supplies semantic anchors for task grounding and geometric free space for navigation. Figure 3 summarizes its construction from exploration data.

1) Target Recognition and Multi-View 3D Fusion: The mapping front end uses text-prompted SAM 3.1 in image mode rather than video tracking. It applies a fixed indoor vocabulary and a confidence threshold specified in Experimental Setup. Each RGB frame returns a 2D proposal, category, confidence,

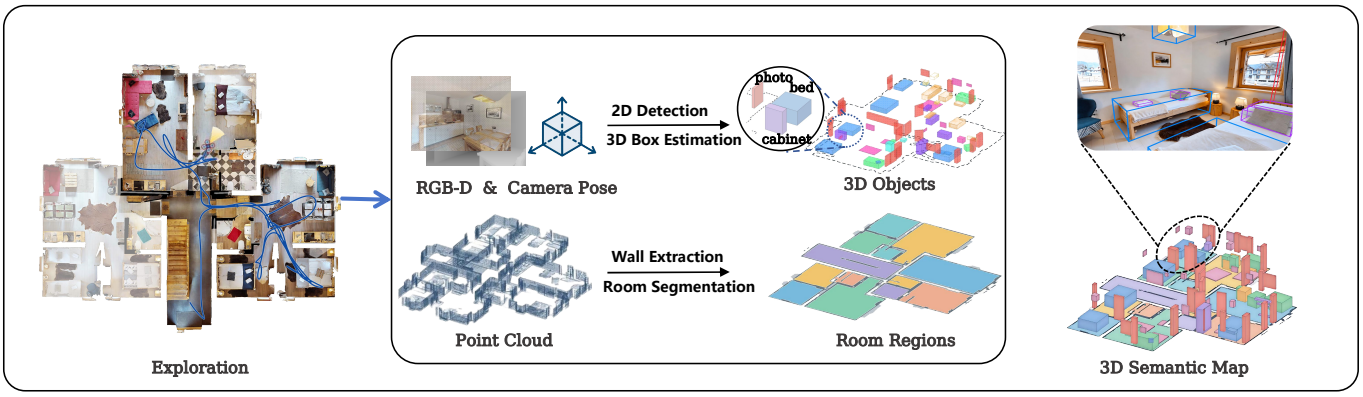

Fig. 3. Construction of the hierarchical semantic–geometric map. RGB-D observations provide object instances, while point-cloud processing yields room regions; these are integrated with exploration data in the 3D map.

and instance mask. The mask preserves foreground depth only inside its proposal footprint and suppresses box-background depth before calibrated RGB-D pixels are lifted into the world frame. Boxer [38] then estimates the oriented 3D instance

$$\mathbf{o}_{t,k} = \big(\mathbf{c}_{t,k}, \mathbf{s}_{t,k}, \mathbf{R}_{t,k}, \ell_{t,k}, p_{t,k}\big), \tag{1}$$

In (1), t is the exact source-frame identity; k is the proposal index; $\mathbf{c}$, $\mathbf{s}$, and $\mathbf{R}$ are the center, size, and orientation, respectively; and ℓ and p are the category and confidence.

Multi-view fusion first applies class-consistent non-maximum suppression and then associates proposals using 3D overlap and semantic agreement. Exact capture timestamps are retained as frame identities, and at most the highest-confidence proposal from each frame contributes to one fused instance. The association thresholds and minimum independent-frame requirement are fixed in Experimental Setup. Category names and semantic IDs are jointly tie-broken deterministically. Let $\mathcal{I}_j$ be the proposals associated with fused instance j, and let $\mathcal{T}_j = \{t \mid \exists k : (t,k) \in \mathcal{I}_j\}$ be their distinct source frames. The confidence-weighted center is computed as

$$\widehat{\mathbf{c}}_j = \frac{\sum_{(t,k)\in\mathcal{I}_j} p_{t,k}\mathbf{c}_{t,k}}{\sum_{(t,k)\in\mathcal{I}_j} p_{t,k}}, \qquad |\mathcal{T}_j| \geq n_{\text{mv}}. \tag{2}$$

The fused instance obtained from (2) becomes the semantic anchor used by room assignment and subsequent language tasks. During online verification, detections enter through the same RGB-D projection interface; known instances use their map IDs, while an unmatched same-category detection becomes a novel online track only after satisfying the minimum-frame and spatial-consistency settings specified in Experimental Setup.

2) Wall- and Observation-Supported Room Partitioning: Room partitioning is performed independently on a horizontal grid for each floor. Wall points extracted from measured geometry are projected into a barrier set $\mathcal{W}$; a bounded local morphological closing only bridges sampling gaps and never hallucinates unknown wall points. The grid resolution and closing neighborhood are fixed in Experimental Setup. Let $\mathcal{G}$ denote the horizontal grid and let $\Omega_0 \subseteq \mathcal{G}$ be the candidate floor domain. To prevent walls or interpolation holes from becoming room evidence, we combine multiple independent

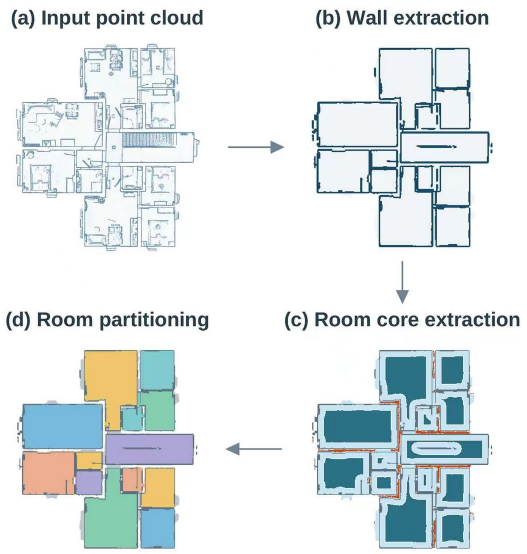

Fig. 4. Room-partitioning pipeline. The input point cloud is processed to extract walls, identify room cores, and produce the final room partition.

support sets: near-horizontal measured surfaces $\mathcal{S}_h$, multi-height free-space observations $\mathcal{S}_f$, multi-frame depth-ray coverage $\mathcal{S}_r$, and recorded trajectory coverage $\mathcal{S}_t$. Each support set is projected onto $\mathcal{G}$ and represented as a set of grid cells. The effective partition domain is defined as

$$\Omega = (\Omega_0 \setminus \mathcal{W}) \cap \big(\mathcal{S}_h \cup \mathcal{S}_f \cup \mathcal{S}_r \cup \mathcal{S}_t\big). \tag{3}$$

According to (3), Ω contains cells that are not blocked by walls and have support from at least one evidence source. A cell without independent support is excluded from the room partition but is not labeled as exterior or occupied; flight feasibility remains determined by the independent 3D occupancy map.

On Ω, we compute the Euclidean distance transform $d(\mathbf{x})$ to the domain boundary. Broad room interiors have larger clearance than doorways, corridors, and room boundaries. We therefore define the interior cores as

$$\mathcal{C} = \{\mathbf{x} \in \Omega \,|\, d(\mathbf{x}) > \tau_c\}, \tag{4}$$

where τ_c is the interior-clearance threshold in (4) and is specified in Experimental Setup. Connected components smaller than $A_{\min}$ are discarded, with $A_{\min}$ fixed in Experimental Setup. Each retained core is a spatial marker rather than a semantic room label. Marker-controlled watershed propagation runs on $-d$ with grid-connected, multi-source priority-queue flooding and is restricted to Ω; it therefore stops at wall barriers and unsupported boundaries, while low-clearance connections become bottlenecks between competing cores. A small enclosed region is merged into its sole labeled neighbor only when its boundary wall support is weak and it does not border unsupported open space. A supported connected component without a marker remains untyped rather than receiving an artificial seed.

Room categories are assigned only after geometry: measured RGB frames whose camera positions lie in a region are paired with that region's fused 3D objects and passed to a multimodal language model acting as a classifier over a fixed set of room categories. The frame-count setting is given in Experimental Setup. Insufficient evidence yields "unknown", and the classifier cannot change the room boundary or occupancy map. Figure 4 summarizes the room-partitioning pipeline, from the input point cloud through wall and room-core extraction to the final partition.

B. Conditional Partial-Order Task Graph

Given a natural-language instruction I, the system constructs a **Conditional Partial-Order Task Graph**

$$\mathcal{G}_T = (\mathcal{V}, \mathcal{E}_{\text{prec}}, \mathcal{R}_{\text{cond}}), \tag{5}$$

where $\mathcal{V}$ denotes language-grounded subtasks, $\mathcal{E}_{\text{prec}}$ specifies mandatory precedence relations, and $\mathcal{R}_{\text{cond}}$ contains observation-dependent activation and skip rules. Each task in (5) stores its action, semantic target, navigation anchor, verification predicate, spatial constraints, room/floor scope, prerequisites, and search policy. The language model produces these fields as structured JSON from the instruction and map inventory. Before execution, the graph is rejected if entity references are invalid, dependencies are inconsistent, or the precedence relation contains a cycle. Figure 5 illustrates the resulting task representation.

For each task, the **navigation anchor** defines the observation region, the **semantic target** specifies the entity to detect, and the **verification predicate** defines the requested evidence. Spatial relations constrain grounding and candidate generation; their satisfaction is recorded geometrically and required by the strict full-task evaluator.

Let $\mathcal{R}_t$ denote tasks whose task-level outcomes have been resolved and $\mathcal{A}_t$ the active pending tasks. The currently executable set is

$$\mathcal{E}_t = \{ i \in \mathcal{A}_t \mid \text{pred}(i) \subseteq \mathcal{R}_t \}. \tag{6}$$

Resolved outcomes are interpreted by $\mathcal{R}_{\text{cond}}$ to activate or skip downstream conditional tasks. Thus, only tasks satisfying (6) enter the planner. Each conditional source is resolved according to its observation policy, either at the first designated viewpoint or after a terminal search outcome. Separating mandatory precedence from conditional activation prevents observation-dependent alternatives from being incorrectly imposed as unconditional order constraints. CPTG thus determines the executable tasks, while JTVP plans their order and observation poses.

C. Task-Conditioned Observation and Motion Modeling

For each executable task i, its navigation anchor and spatial constraints define a task-specific 3D observation region, such as an object, support surface, surrounding area, lower region, or relation-defined area.

The candidate observation pose indexed by k is $\mathbf{q}_i^k = (x_i^k, y_i^k, z_i^k, \psi_i^k)$, where (x_i^k, y_i^k, z_i^k) is the camera position and ψ_i^k is yaw. Candidates are sampled at multiple distances, azimuths, and feasible heights, with yaw oriented toward the observation region; vertical visibility is handled by candidate height and the camera field of view.

Let $\mathcal{P}_i$ denote samples from the observation region and $\mathcal{V}_i(\mathbf{q}) \subseteq \mathcal{P}_i$ the samples visible from $\mathbf{q}$ under camera field-of-view and line-of-sight constraints. We define the geometric coverage as

$$Q_i(\mathbf{q}) = \frac{|\mathcal{V}_i(\mathbf{q})|}{\max\{1, |\mathcal{P}_i|\}}. \tag{7}$$

Candidates violating occupancy, flight-height, spatial-relation, or required-visibility constraints are discarded. The remaining poses are scored using the coverage in (7):

$$\begin{aligned} C_i^{\text{view}}(\mathbf{q}) = \beta_d \frac{|d - d_i^{\text{pref}}|}{\ell_0} + \beta_Q \big[1 - Q_i(\mathbf{q})\big] \\ + \beta_z \frac{|z - \bar{z}_i|}{\ell_0} + \beta_r \frac{d_i^{\text{ref}}(\mathbf{q})}{\ell_0} + \beta_b \frac{d_i^{\text{between}}(\mathbf{q})}{\ell_0}. \end{aligned} \tag{8}$$

Here, d is the sampled horizontal standoff from the anchor's bounding footprint, d_i^{pref} is its preferred value, z is the candidate height, and $\bar{z}_i$ is the mean observation-region height. Let $a(\mathbf{q})$ identify the anchor that generated a candidate, $\mathbf{c}_{a,xy}$ its horizontal center, $\mathbf{x}(\mathbf{q}) = (x, y)$ the candidate's horizontal position, and $\mathcal{B}_i$ the instruction's reference objects. The relation terms, measured in meters, are

$$\begin{aligned} d_i^{\text{ref}}(\mathbf{q}) &= \min_{b \in \mathcal{B}_i} \|\mathbf{c}_{a(\mathbf{q}),xy} - \mathbf{c}_{b,xy}\|_2, \\ d_i^{\text{between}}(\mathbf{q}) &= \text{dist}\big(\mathbf{x}(\mathbf{q}), [\mathbf{c}_{b_1,xy}, \mathbf{c}_{b_2,xy}]\big). \end{aligned} \tag{9}$$

The first term is zero when $\mathcal{B}_i$ is empty; it compares alternative anchors and is constant across views of the same anchor. The second term is used only for a "between" relation with two distinct reference centers and is the distance to their closed line segment. Let $\mathbf{h} = \mathbf{c}_{b_2,xy} - \mathbf{c}_{b_1,xy}$ and $u = (\mathbf{x}(\mathbf{q}) - \mathbf{c}_{b_1,xy})^{\mathsf{T}}\mathbf{h}/\|\mathbf{h}\|_2^2$ be the normalized projection parameter; the hard filter requires $u \in [-0.05, 1.05]$ and segment distance at most $\max(0.75\,\text{m}, d_{i,\max}/2)$, where $d_{i,\max}$ is the maximum sampling standoff. Coincident reference centers are rejected. Other inapplicable terms are zero. The reference length ℓ_0 makes (8) dimensionless. Hard feasibility is enforced before scoring, so a low soft cost cannot compensate for an unsafe or unobservable pose. All weights are nonnegative and given in Section V-A. The retained set $\mathcal{Q}_i = \{\mathbf{q}_i^1, \ldots, \mathbf{q}_i^{K_i}\}$ preserves diverse feasible viewpoints.

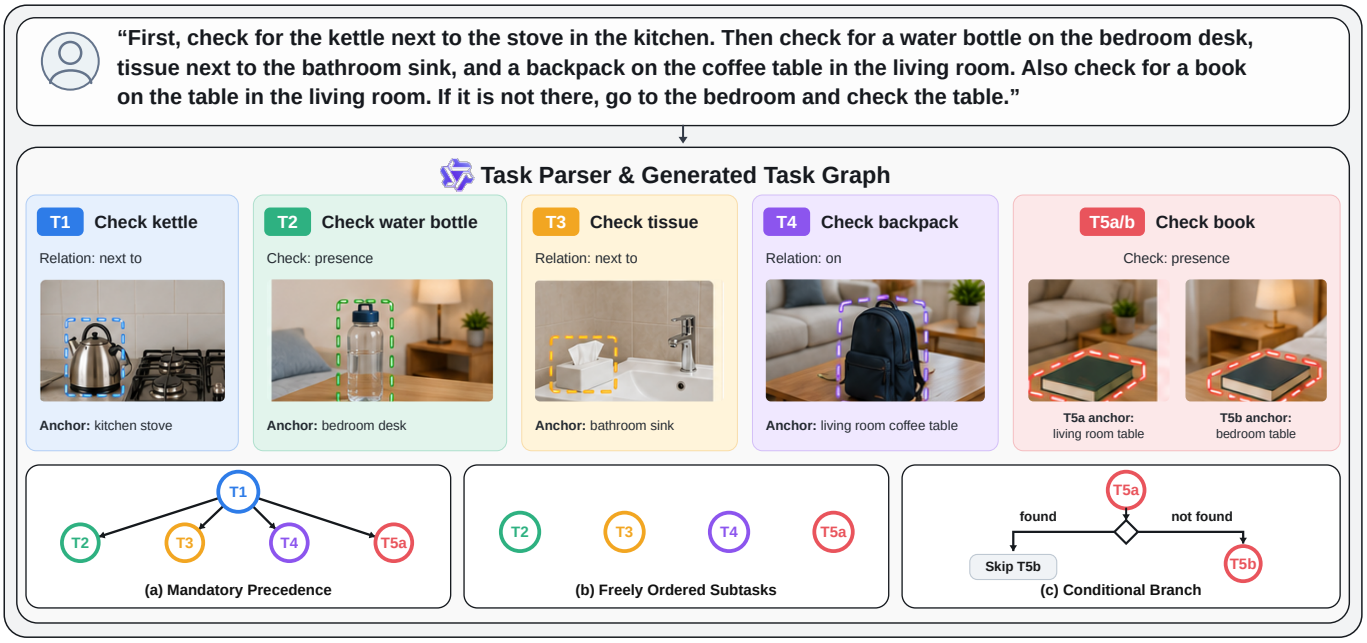

Fig. 5. From a natural-language instruction to a CPTG. The task parser constructs task nodes with navigation anchors, semantic targets, and verification predicates; precedence and conditional edges encode mandatory ordering and observation-dependent activation.

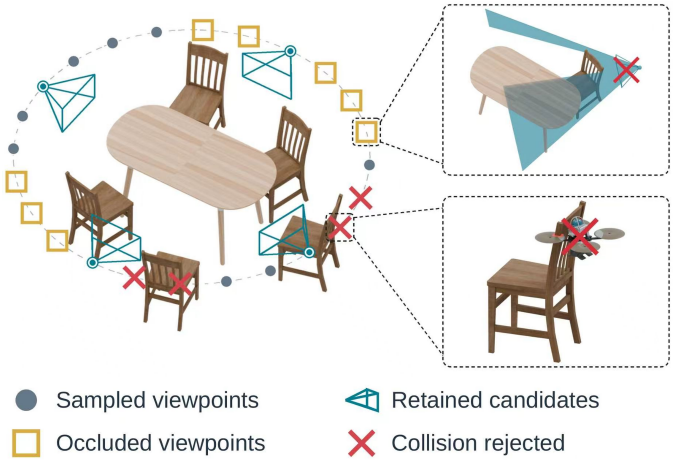

Fig. 6. Viewpoint sampling and filtering for a dining-table target. Sampled poses are rejected when occluded or collision-prone; the remaining feasible views are retained for observation.

Figure 6 illustrates the sampling and feasibility-filtering process.

For candidate poses $\mathbf{q}_a$ and $\mathbf{q}_b$, a 3D A* query returns a collision-free polyline $P_t(\mathbf{q}_a, \mathbf{q}_b) = \{\mathbf{p}_1, \ldots, \mathbf{p}_N\}$ when the transition is reachable. Let $\Delta\psi_{ab} = \mathrm{wrap}(\psi_b - \psi_a)$. The path length and motion-time proxy are defined, respectively, as

$$L_t(\mathbf{q}_a, \mathbf{q}_b) = \sum_{u=1}^{N-1} \|\mathbf{p}_{u+1} - \mathbf{p}_u\|_2, \tag{10a}$$

$$\widehat{T}_t(\mathbf{q}_a, \mathbf{q}_b) = \max\left\{ \frac{L_t(\mathbf{q}_a, \mathbf{q}_b)}{v_{\mathrm{nom}}}, \frac{|z_b - z_a|}{v_z}, \frac{|\Delta\psi_{ab}|}{\omega_\psi} \right\}. \tag{10b}$$

Here, $v_{\mathrm{nom}} > 0$ is a nominal speed along the three-dimensional polyline in (10b), while $v_z > 0$ and $\omega_\psi > 0$ set nominal climb and yaw rates. The wrap operator returns an angle in $[-\pi, \pi)$. The maximum models concurrent translation and yaw as a ranking heuristic, not the duration of the optimized trajectory. Cross-floor detours and intermediate ascents or descents contribute to L_t; the second term accounts only for net altitude change. Acceleration, intermediate vertical-rate constraints, and observation dwell are handled during execution rather than predicted by this proxy. Unreachable transitions have $L_t = \widehat{T}_t = +\infty$.

Figure 7 shows A*-guided trajectory refinement after a negative observation, including route replanning and trajectory optimization.

D. Receding-Horizon Joint Task–Viewpoint Planning

At each planning cycle, the planner selects a local task window $\mathcal{W}_t \subseteq \mathcal{E}_t$ with $|\mathcal{W}_t| \le H$. Tasks are prioritized by their best candidate's current-position distance plus terminal cost, with task IDs breaking ties; recovery can focus the window on the task under reinspection. Every predecessor of a task in $\mathcal{E}_t$ is already resolved. Thus, no additional precedence constraint is needed within this window; newly enabled successors enter a later planning cycle.

Candidate poses are grouped by physical location and prioritized using current-pose distance, motion-time estimate, and observation score. The retained set $\widetilde{\mathcal{Q}}_{i,t} \subseteq \mathcal{Q}_i$ maintains representative viewpoints across semantic locations and is progressively augmented when alternatives are needed.

Given $m = |\mathcal{W}_t| > 0$, let $\Pi(\mathcal{W}_t)$ be all permutations of the window and $\mathcal{K}_{i,t} = \{k : \mathbf{q}_i^k \in \widetilde{\mathcal{Q}}_{i,t}\}$. JTVP jointly selects $\pi \in \Pi(\mathcal{W}_t)$, visiting each task exactly once, and one observation pose $\mathbf{q}_r = \mathbf{q}_{\pi_r}^{k_r}$ for each selected task, with $\mathbf{q}_0 = \mathbf{q}_t^{\mathrm{robot}}$. Let

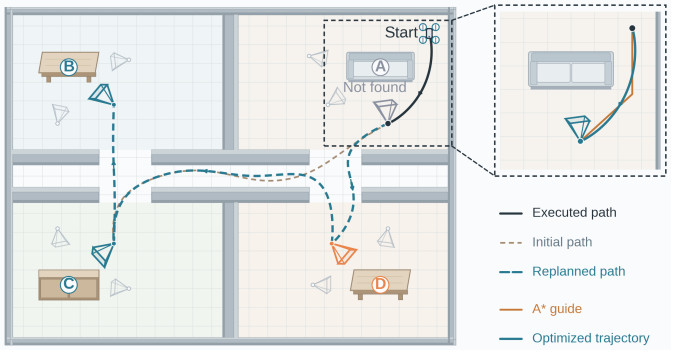

Fig. 7. Joint task–viewpoint planning and trajectory refinement. After an observation returns no target, the planner replans and produces a collision-free trajectory guided by A*.

$\iota(\mathbf{q})$ denote the task associated with candidate $\mathbf{q}$. We define the dimensionless transition score as

$$c_t(\mathbf{q}_a, \mathbf{q}_b) = \eta_L \frac{L_t(\mathbf{q}_a, \mathbf{q}_b)}{\ell_0} + \eta_T \frac{\widehat{T}_t(\mathbf{q}_a, \mathbf{q}_b)}{\tau_0} + \eta_V C^{\text{view}}_{\iota(\mathbf{q}_b)}(\mathbf{q}_b), \tag{11}$$

where τ_0 is a reference time. Using (11), JTVP solves

$$\begin{aligned} \underset{\pi,\mathbf{k}}{\text{minimize}} \quad & \sum_{r=1}^{m} c_t(\mathbf{q}_{r-1}, \mathbf{q}_r) \\ \text{subject to} \quad & \pi \in \Pi(\mathcal{W}_t), \quad k_r \in \mathcal{K}_{\pi_r,t}, \\ & \mathbf{q}_r = \mathbf{q}^{k_r}_{\pi_r}, \quad r = 1, \dots, m, \\ & L_t(\mathbf{q}_{r-1}, \mathbf{q}_r) < \infty, \quad r = 1, \dots, m. \end{aligned} \tag{12}$$

The joint planning problem in (12) is solved using subset dynamic programming. Let $D(S, i, k)$ be the minimum cost of visiting exactly $S \subseteq \mathcal{W}_t$ and ending at $\mathbf{q}^k_i$, where $i \in S$ and $k \in \mathcal{K}_{i,t}$. Invalid or unreachable states have value $+\infty$, as do unreachable transition scores. For nonempty S and $j \in \mathcal{W}_t \setminus S$, the initialization, recurrence, and terminal cost are

$$D(\{i\}, i, k) = c_t(\mathbf{q}_0, \mathbf{q}^k_i), \tag{13a}$$

$$D(S \cup \{j\}, j, \ell) = \min_{i \in S,\, k \in \mathcal{K}_{i,t}} \left[D(S, i, k) + c_t(\mathbf{q}^k_i, \mathbf{q}^\ell_j) \right], \tag{13b}$$

$$J^\star_t = \min_{i \in \mathcal{W}_t,\, k \in \mathcal{K}_{i,t}} D(\mathcal{W}_t, i, k). \tag{13c}$$

The recurrence is evaluated for $\ell \in \mathcal{K}_{j,t}$; a minimum over an empty set is $+\infty$. An empty window yields an empty plan with zero cost: execution finishes only if all required outcomes are resolved, otherwise it reports unresolved tasks. Empty candidate sets or an infinite terminal cost trigger expansion from representative views to the full available pool. If no feasible plan is found, planning terminates with failure rather than marking the task verified. For a finite terminal cost, backtracking through (13) returns the optimal order and poses for the selected window and candidate sets. Only the first visit is executed before replanning from the updated robot pose, task state, candidate set, and semantic evidence. This receding-horizon policy preserves the efficiency of multi-task lookahead without committing later visits to assumptions that subsequent observations may invalidate. With $K = \max_{i \in \mathcal{W}_t} |\mathcal{K}_{i,t}|$ and cached transition costs, the subset DP uses $O(2^m m^2 K^2)$ time and $O(2^m m K)$ memory. Since $m \le H$, the exponential dependence is limited to the local window. Task screening, candidate generation, and A* queries still depend on the available tasks and map size; retaining views per location does not impose a global constant K when additional locations are admitted. The 15-subtask timing in Section V-B measures the complete planner under this policy.

E. Safe-Corridor-Constrained Trajectory Optimization

After JTVP selects the next task and observation pose, the trajectory optimizer connects the discrete A* route to the start and goal, constructs geometric safe corridors, and optimizes one continuous trajectory. Safe corridors $\mathcal{K}_r$ are generated from measured point-cloud clearance and Configuration-Space Iterative Regional Inflation (CIRI) [39] geometric seeds, and the resulting piecewise polynomial trajectory is optimized with MINCO [40]. Unknown voxels are treated as blocked, narrow passages are represented by local guide centerlines and centered half-space constraints, and adjacent corridor intersections are checked to prevent a smoothed curve from cutting across a wall.

For R trajectory pieces, the position on piece r is a seventh-degree polynomial,

$$\mathbf{p}_r(\tau) = \sum_{h=0}^{7} \mathbf{a}_{r,h} \tau^h, \qquad \tau \in [0, T_r], \quad T_r > 0. \tag{14}$$

MINCO determines the coefficients from internal interpolation points $\mathbf{U} = (\mathbf{u}_1, \dots, \mathbf{u}_{R-1})$, durations $\mathbf{T}$, and prescribed endpoint position, velocity, acceleration, and jerk. Adjacent pieces share derivatives through order three. Internal points are parameterized in the corresponding corridor intersections. Writing $\mathbf{p} = \mathcal{M}(\mathbf{U}, \mathbf{T})$ for this minimum-snap parameterization, the optimization is

$$\begin{aligned} \min_{\mathbf{U},\, \mathbf{T} > \mathbf{0}} \quad & \sum_{r=1}^{R} \int_0^{T_r} \|\mathbf{p}^{(4)}_r(\tau)\|_2^2 \, d\tau + \rho \sum_{r=1}^{R} T_r \\ & + J_{\text{corr}}(\mathbf{p}, \mathbf{T}) + J_{\text{dyn}}(\mathbf{p}, \mathbf{T}), \end{aligned} \tag{15}$$

where $\rho > 0$ trades trajectory duration against snap energy. Let corridor $\mathcal{K}_r$ have normalized half-space constraints $\mathbf{n}^\mathsf{T}_{r,f}\mathbf{p} + b_{r,f} \le 0$. The soft penalties have the explicit form

$$\begin{aligned} J_{\text{corr}} &= \lambda_c \sum_{r,f} \int_0^{T_r} \phi_\mu(\mathbf{n}^\mathsf{T}_{r,f}\mathbf{p}_r + b_{r,f}) \, d\tau, \\ J_{\text{dyn}} &= \sum_{r} \sum_{d \in \mathcal{D}} \lambda_d \int_0^{T_r} \phi_\mu(g_d(\mathbf{p}_r, \tau)) \, d\tau. \end{aligned} \tag{16}$$

Here, $\mathcal{D}$ indexes enabled dynamic penalties. Translational penalties use $g_d = \|\mathbf{p}^{(d)}_r\|_2^2 - b_d^2$ for derivative orders $d = 1, 2, 3$ and limits b_d on speed, acceleration, and jerk. Enabled body-rate and thrust penalties use the corresponding differential-flatness quantities and configured bounds. The nonnegative weights carry the units needed to combine these terms. For smoothing parameter $\mu > 0$, the implementation uses the one-sided penalty

$$\phi_\mu(x) = \begin{cases} 0, & x \le 0, \\ (\mu - x/2)(x/\mu)^3, & 0 < x < \mu, \\ x - \mu/2, & x \ge \mu, \end{cases} \tag{17}$$

Penalty integrals use trapezoidal quadrature with at least 40 subdivisions per piece. Endpoint states, MINCO continuity, positive durations, and internal-point corridor membership are imposed by parameterization; whole-piece corridor and dynamic violations are penalized during optimization and checked before acceptance. Optional guide-attraction and regional-height terms supplement this formulation. Snap limits are checked on the resulting trajectory rather than introduced as a separate snap-limit penalty.

Yaw is generated separately from the position polynomial. A direction-following reference is blended toward the required observation yaw, unwrapped, and sampled at intervals no greater than $0.01\,\mathrm{s}$. A rate-limited profile with prescribed start and terminal yaw is then formed and linearly interpolated. A phase too short to satisfy the yaw-rate bound is rejected.

Trajectory validation uses a maximum temporal step of $0.03\,\mathrm{s}$ and a maximum speed-based spatial step of $0.01\,\mathrm{m}$. For a validation interval of duration Δ with speed bound $v_{\max}^{\mathrm{poly}}$ obtained from the polynomial, the distance to the fixed collision point cloud satisfies

$$\min_{\tau} d_{\mathrm{cloud}}(\mathbf{p}(\tau)) \geq \min_{j} d_{\mathrm{cloud}}(\mathbf{p}(\tau_j)) - \tfrac{1}{2} v_{\max}^{\mathrm{poly}} \Delta. \tag{18}$$

This follows from the 1-Lipschitz property of distance and sampling both ends of each interval. The reserved clearance must exceed the collision margin. This bound concerns the supplied collision geometry; finite sampling alone is not a continuous-time guarantee. Corridor, regional-height, derivative, and yaw-rate checks must also pass before acceptance. Replanning retains the committed prefix and regenerates the remaining route.

F. Observation-Driven Semantic Revision and Recovery

At the selected observation pose, RGB observations acquired during motion contribute semantic evidence, while terminal RGB-D observations provide open-vocabulary detections, depth-based 3D projections, and geometric verification cues. A positive online verification advances the task state. A negative terminal observation is handled according to the task-specific not-found policy: it may trigger another local viewpoint, bounded semantic recovery, a terminal absence result, or an observation-dependent branch.

Open-vocabulary detections are lifted into the global frame using depth and camera pose. Same-category observations are associated with mapped instances through spatial consistency, while unassociated detections are accumulated across views for subsequent scene-graph revision.

For tasks configured for semantic recovery, LogicFly distinguishes three levels of negative evidence. A **pose-level miss** removes the attempted candidate while keeping the task active. **Region-level exhaustion** invokes the language model with the target, mapped room/object inventory, and failed search anchors to propose alternative room–functional-region hypotheses grounded in the hierarchy; the hypotheses must reference existing room and object IDs and pass deterministic validation before becoming geometric observation regions and are ranked using semantic relevance, coverage, and travel cost. A **task-level negative outcome** is produced when the task-specific stopping condition is reached, after which CPTG updates downstream logic. Within semantic recovery, newly covered portions of an unproductive region reduce its belief, shifting search toward alternative hypotheses.

For a semantic recovery region r, let $\mathcal{P}_r$ be its fixed sample set and $\mathcal{O}_{r,n}$ the samples already covered before verification n, initialized as $\mathcal{O}_{r,0} = \varnothing$. With visible samples $\mathcal{V}_r(\mathbf{q}_n)$, define

$$\begin{aligned} \delta_{r,n} &= \frac{|\mathcal{V}_r(\mathbf{q}_n) \setminus \mathcal{O}_{r,n}|}{\max(1, |\mathcal{P}_r|)}, \\ \mathcal{O}_{r,n+1} &= \mathcal{O}_{r,n} \cup \mathcal{V}_r(\mathbf{q}_n). \end{aligned} \tag{19}$$

The task-local confidence $b_{r,0}$ is initialized from semantic relevance: high, medium, low, and fallback hypotheses receive 1.0, 0.6, 0.3, and 0.15. Registration preserves an existing region's accumulated state. If $y_n \in \{0, 1\}$ indicates successful target verification and $Q_r(\mathbf{q}_n)$ is geometric view quality, the update is

$$b_{r,n+1} = \begin{cases} 1, & y_n = 1, \\ b_{r,n}\big[1 - \alpha \delta_{r,n} Q_r(\mathbf{q}_n)\big], & y_n = 0, \end{cases} \tag{20}$$

with detector reliability $\alpha = 0.80$. These scores remain in $[0, 1]$ and are not normalized across regions or interpreted as a posterior probability distribution. Repeated inspection of already covered samples gives $\delta_{r,n} = 0$, avoiding repeated confidence reduction from the same coverage.

Recovery views replace the ordinary terminal score by

$$C_{r,n}^{\mathrm{rec}}(\mathbf{q}) = 1.5(1 - b_{r,n}) + 0.6\widehat{c}_r + 0.8[1 - Q_r(\mathbf{q})]. \tag{21}$$

Here, $\widehat{c}_r$ is the min–max normalized coverage-route cost across proposed regions, set to zero if all costs coincide. The route estimate greedily selects up to four views by newly covered samples, sums their A* travel lengths, and adds $2\,\mathrm{m}$ times the uncovered fraction; an unreachable estimate receives a $10^3\,\mathrm{m}$ sentinel cost. Thus, confidence and coverage affect terminal quality, while travel from the current pose remains in (11).

The recovery procedure is: (1) verify the selected view and update (19)–(20); (2) remove a missed view and retry unused local candidates; (3) on local exhaustion, admit permitted semantic-region hypotheses and rescore using (21); and (4) resolve the task when verification succeeds or its declared observation/search policy terminates. The implementation allows at most three alternative hypotheses per request and four distinct locations per task. A first-viewpoint branch is decided at that designated observation rather than delayed for exhaustive search. Exhausting this budget means "not found within the prescribed search policy", not proof of global nonexistence; only a matching task predicate accepts that negative outcome. An unresolved positive-search goal or an episode timeout remains a failure.

Novel detections are accumulated online through spatially consistent multi-view fusion and materialized in the updated scene graph after execution. Thus, task states and semantic-region beliefs can affect replanning within the current execution, whereas newly created object instances support later use of the revised map.

After each terminal verification, the system updates the task state, candidate poses, and semantic search prior, recomputes

$\mathcal{E}_t$, and invokes JTVP from the current robot pose. When a task reaches a task-level outcome, CPTG additionally updates $\mathcal{R}_t$ and evaluates $\mathcal{R}_{\text{cond}}$ to activate or skip downstream conditional tasks. This closes the loop among task logic, 3D observation, geometric motion, and online semantic evidence.

V. Experiments

We evaluate LogicFly through simulation, ablation studies, and real-world UAV experiments, covering benchmark performance, the contributions of CPTG, JTVP, and semantic recovery, and practical deployment.

A. Experimental Setup

Evaluation benchmark. We evaluate all methods in the Habitat simulator [41] on six HM3D [42] scenes: 00070, 00166, 00273, 00290, 00337, and 00409. The benchmark contains 152 language instructions, including 61 short single-target instructions, 60 long-horizon instructions confined to a single floor, and 31 long-horizon cross-floor instructions. Using the valid initial-state seeds and two repeated trials, each method is evaluated over 854 episodes. The benchmark covers ordered execution, conditional branches, target absence, failure recovery, spatial relations, and instance disambiguation.

Evaluation metrics. We report task Success Rate (SR), Success weighted by Path Length (SPL), and Completion Time (CT). An episode is considered successful only when all required task outcomes are completed while satisfying the instruction-level ordering, branching, spatial, recovery, and disambiguation constraints. Reaching the correct objects without satisfying the complete instruction is counted as a failure. For an episode e with a defined geometric reference route, the reported path-efficiency metric is

$$\text{SPL}_e = S_e \frac{L_e^{\text{ref}}}{\max\{L_e^{\text{ref}}, L_e, \varepsilon_L\}}, \qquad \varepsilon_L = 10^{-9}\,\text{m}, \tag{22}$$

where $S_e \in \{0, 1\}$ is strict task success and L_e is the sum of three-dimensional distances along the recorded trajectory. The evaluator constructs L_e^{ref} retrospectively using the required tasks selected by the recorded conditional-source observation, retaining instruction precedence and branch ordering. If that source was never observed, all potentially required tasks remain in the reference problem. It uses the same voxel geometry, a finite generated candidate set, A* transition costs, and the joint planner's weighted objective; L_e^{ref} is the length of the resulting route, not a certified shortest path in continuous space or a pure path-length optimum. This reference construction is separate from the online policy and does not provide future branch outcomes to the robot.

For a required target-absence judgment, the frozen search-anchor center is used as the reference endpoint, with the planner's nearest-free-voxel adjustment. These episodes are therefore included when that anchor yields a valid reference route. No unique endpoint or no feasible reference route makes SPL undefined. For applicable episodes, failures and timeouts contribute zero; the denominator is the number of applicable episodes, not the number of successful episodes. The recorded LogicFly results have 854 applicable episodes in Table I, including 54 target-absence episodes, and all 546 multi-target episodes are applicable in Table II. Under the implemented numerical convention, $L_e^{\text{ref}} = 0$ yields zero even if $L_e = 0$; otherwise (22) lies in $[0, 1]$. Group scores are arithmetic means over their applicable episodes.

Baselines. We compare four map-based and spatial-reasoning methods. OneMap [22] queries a real-time open-vocabulary feature map for multi-object navigation. MapGPT [6] uses map-guided LLM prompting and adaptive route planning. SpatialGPT [7] applies spatial chain-of-thought reasoning over structured memory. SpatialNav [28] uses object-relation scene graphs for zero-shot navigation.

Implementation details. LogicFly first compiles each instruction into a conditional and precedence-constrained task graph, which is subsequently executed by the proposed task-graph runner. Reusable semantic maps are constructed from registered RGB-D observations. The mapping front end uses SAM 3.1 in image mode with a fixed vocabulary of 30 indoor object categories and a confidence threshold of 0.50. Video tracking is disabled. Boxer lifts the accepted masks into 3-D and performs multi-view instance fusion using an IoU threshold of 0.20, an instance-confidence threshold of 0.45, and a minimum of four observations. During task execution, online object evidence is obtained from calibrated RGB-D observations using open-vocabulary recognition and is projected into the world frame using the estimated camera pose and aligned depth.

Room partitioning uses a $0.05\,\text{m}$ horizontal grid, a 3×3 wall-closing neighborhood, an interior-clearance threshold of $\tau_c = 0.45\,\text{m}$, and a minimum core area of $A_{\min} = 0.04\,\text{m}^2$. Horizontal-surface support requires $|n_z| \geq 0.90$ and a projected support distance within $0.12\,\text{m}$. Free-space support requires observations at least two of four heights, 0.45, 0.85, 1.25, and $1.65\,\text{m}$ above the floor, with a $0.09\,\text{m}$ proximity tolerance. Depth-ray support requires at least three source frames, and trajectory support uses a $0.15\,\text{m}$ radius. A region can merge into its sole neighbor only when the boundary wall fraction is below 0.20 and the boundary does not touch unsupported open space. Room classification uses up to two selected RGB views and a confidence threshold of 0.65; insufficient evidence yields an unknown label. Online fusion associates same-category novel detections within $0.75\,\text{m}$ and requires at least two supporting observations before materializing a new instance.

The joint task-and-viewpoint planner retains up to three representative viewpoints per location and uses a receding planning horizon of $H = 3$ tasks. In (8), $(\beta_d, \beta_Q, \beta_z, \beta_r, \beta_b) = (1, 1, 0.10, 0.15, 0.15)$. In (11), $(\eta_L, \eta_T, \eta_V) = (1, 0.25, 2)$, with reference scales $\ell_0 = 1\,\text{m}$ and $\tau_0 = 1\,\text{s}$. The nominal path speed, climb speed, and yaw rate are $1.0\,\text{m/s}$, $0.5\,\text{m/s}$, and $1.0\,\text{rad/s}$, respectively. A collision safety margin of $0.10\,\text{m}$ is enforced during trajectory generation in both simulated and real-world experiments. All language-model components use Qwen3.8-Flash. Semantic recovery generates at most three alternative hypotheses per request and permits at most four distinct location hypotheses per task, with up to three viewpoints retained per location. Local reinspection exhausts the finite candidate pool before advancing to another permitted

TABLE I

BENCHMARK RESULTS GROUPED BY TASK DIFFICULTY AND FLOOR SCOPE. THE CLEAN SCHEDULE CONTAINS 270 SINGLE-FLOOR AND 38 CROSS-FLOOR SINGLE-TARGET EPISODES, AND 364 SINGLE-FLOOR AND 182 CROSS-FLOOR MULTI-TARGET EPISODES. SR AND SPL ARE PERCENTAGES; CT IS AVERAGED OVER SUCCESSFUL EPISODES. DASHES OMIT SPL AND CT WHEN NO EPISODE SUCCEEDS. ONEMAP USES ITS ORDERED-CATEGORY PROJECTION. BOLD DENOTES THE BEST RESULT IN EACH TASK GROUP.

Task type	Method	Single-floor			Cross-floor		
		SR↑ (%)	SPL↑ (%)	CT↓ (s)	SR↑ (%)	SPL↑ (%)	CT↓ (s)
Single-target	SpatialNav	8.15	5.89	22.35	0.00	–	–
	SpatialGPT	10.37	2.69	115.39	0.00	–	–
	MapGPT	27.78	14.35	42.95	18.42	13.08	81.57
	OneMap	52.59	35.45	77.32	21.05	20.30	49.88
	LogicFly	**97.78**	**87.28**	**8.19**	**94.74**	**86.03**	**8.99**
Multi-target	MapGPT	1.10	0.73	96.02	0.00	–	–
	OneMap	12.36	8.16	150.13	6.59	6.59	90.59
	LogicFly	**96.70**	**82.49**	**10.33**	**96.70**	**81.45**	**14.70**

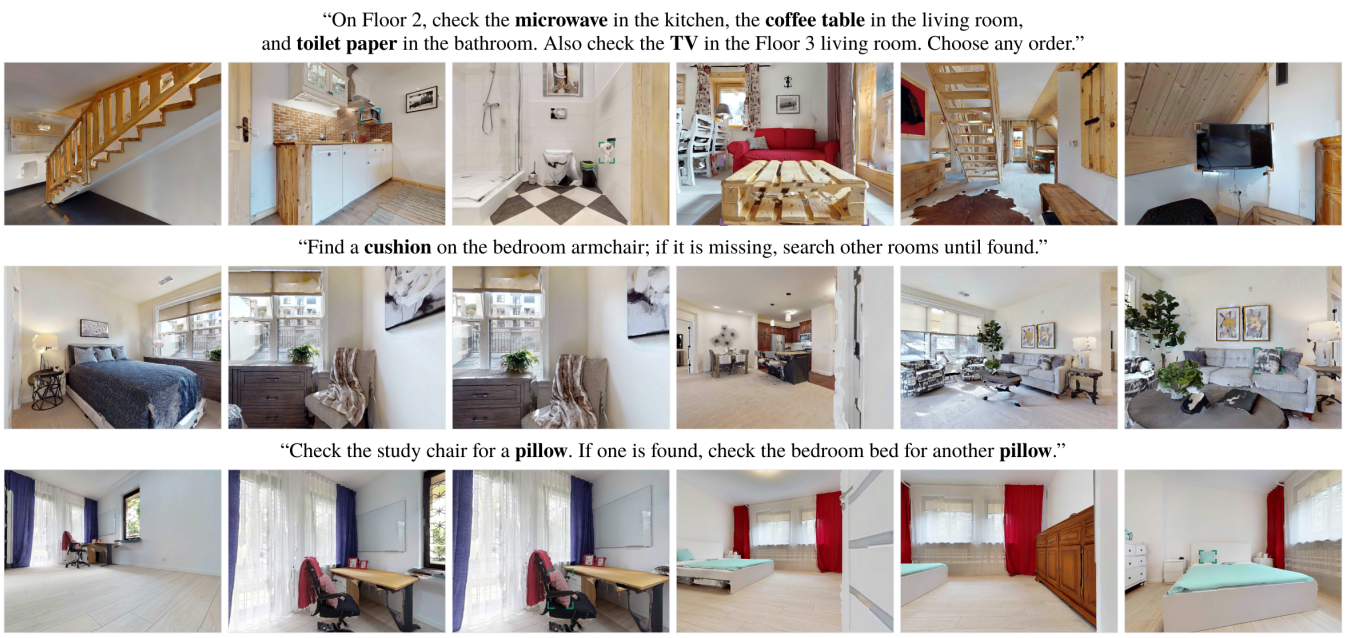

Fig. 8. Examples of long-instruction execution in Habitat. Each row pairs a natural-language instruction with the corresponding sequence of first-person observations, covering freely ordered subtasks and observation-dependent search or verification.

location. High-, medium-, and low-relevance region priors are 1.0, 0.6, and 0.3, respectively; the fallback prior is 0.15. Negative-evidence updates use detector reliability 0.80 and newly covered samples, with scoring weights of 1.5, 0.6, and 0.8 for belief deficiency, normalized coverage cost, and view-quality deficiency, respectively. Task-specific branch and not-found policies determine whether a negative observation immediately resolves a task or invokes bounded recovery. All perception, planning, and recovery parameters are fixed across scenes and episode repetitions.

Runtime protocol. Each evaluation process is subject to a fixed time limit of 240 s. A timeout or execution error is counted as a failed episode. For LogicFly, CT is measured using its native completion timestamp and excludes one-time map preparation. For the comparison methods, CT is measured from process startup to task completion. CT is averaged over successful episodes only.

B. Comparison with Baseline Methods in Simulation

We compare LogicFly with HM3D adaptations of OneMap [22], MapGPT [6], SpatialGPT [7], and SpatialNav [28]. All methods run on the same 854-episode clean schedule. OneMap retains its native ordered-category projection, which omits relations, conditional branches, verified absence, recovery, and return constraints. All baselines and LogicFly are evaluated using the same frozen strict full-task evaluator. SpatialGPT and SpatialNav do not successfully complete any cross-floor episode, resulting in an SR of 0%; therefore, their remaining cross-floor metrics are not reported.

As shown in Table I, LogicFly outperforms the compared methods across target counts and floor scopes. On multi-target tasks, it maintains 96.70% SR on both floor scopes, with SPL

of 82.49% and 81.45% and CT of 10.33 s and 14.70 s for single-floor and cross-floor episodes, respectively. SpatialGPT and SpatialNav complete no multi-target episodes and are omitted from that block. Overall, LogicFly completes 828/854 episodes (96.96% strict full-task SR, 0.8394 SPL).

Execution with more subtasks. In an additional cross-floor simulation in scene 00337, LogicFly completes all 15 subtasks of a single natural-language instruction in one mission. With the same fixed planning horizon $H = 3$, language parsing and native navigation share a 240 s deadline; the complete run takes 148.96 s, including 59.24 s for parsing and 89.72 s for navigation. The 95th-percentile planning-call latency is 7.95 s. This run demonstrates completion beyond the planning-window size without a timeout; it is a single-run scalability demonstration rather than an estimate of the maximum supported task count.

Figure 8 presents a qualitative execution example in a three-floor Habitat environment. LogicFly completes the cross-floor route and all required subtasks, with terminal views providing visual evidence for each successful outcome.

All baselines retain their native planning, action, and stopping interfaces. OneMap receives the protocol-defined ordered-category projection, whereas MapGPT, SpatialGPT, and SpatialNav receive the frozen task specification through their native interfaces. Timeouts and algorithm failures remain unsuccessful terminal outcomes.

C. Ablation Studies

CPTG and JTVP. We evaluate CPTG and JTVP on the 546 multi-target episodes from the frozen six-scene HM3D schedule, covering all multi-target task types. The analysis contains 1638 end-to-end records, reusing the corresponding native benchmark runs for the full method. All three variants use outer-run completion times for CT, including the reused records.

All variants share the native Habitat runner, online open-vocabulary verification, strict evaluator, scene representations, start poses, motion constraints, and termination settings. *w/o CPTG* removes task dependencies, conditional rules, and task-conditioned candidate generation, retaining target labels, coarse room/floor scopes, generic observation constraints, and fixed language order. *w/o JTVP* retains CPTG but fixes task order and uses only each task's primary viewpoint. *LogicFly* jointly optimizes task ordering and viewpoints. The evaluator enforces required outcomes and ordering, forbids visits to skipped branches, and checks spatial/reference constraints.

Table II shows that removing CPTG reduces SR to 64.84% and SPL to 58.45%, while increasing successful-task CT to 35.63 s. Failures to enforce precedence and observation-dependent branches lead to incorrect ordering, unnecessary visits, and repeated recovery, highlighting CPTG's contribution to full-task correctness. Retaining CPTG with fixed order and primary viewpoints completes 528/546 episodes (96.70% SR) without infrastructure failures, with 76.82% SPL and 12.62 s CT. JTVP preserves this SR while raising SPL by 5.33 percentage points to 82.15% and reducing CT to 11.79 s (approximately 6.6%). Thus, CPTG primarily supports instruction correctness, while JTVP improves path and time efficiency. The 30 w/o CPTG timeouts remain in the 546-episode SR and SPL denominators, with zero SPL.

TABLE II
ABLATION RESULTS ON THE MULTI-TARGET SUBSET OF 546 EPISODES. SR AND SPL ARE PERCENTAGES. CT USES THE SAME OUTER-RUN TIMING FIELD FOR ALL VARIANTS AND IS AVERAGED OVER SUCCESSFUL EPISODES.

Variant	SR↑ (%)	SPL↑ (%)	CT↓ (s)
w/o CPTG	64.84	58.45	35.63
w/o JTVP	**96.70**	76.82	12.62
LogicFly	**96.70**	**82.15**	**11.79**

TABLE III
SEMANTIC-RECOVERY ABLATION ON 10 CROSS-REGION TASKS, WITH 60 EPISODES PER METHOD. SR AND SPL ARE PERCENTAGES; CT IS AVERAGED OVER SUCCESSFUL EPISODES. A DASH DENOTES NO SUCCESSFUL EPISODE.

Variant	SR↑ (%)	SPL↑ (%)	CT↓ (s)
w/o recovery	0.00	0.00	–
LogicFly	**60.00**	**42.43**	12.78

Semantic recovery. We further evaluate 10 cross-region target-search tasks across the same six scenes, covering both same-floor and cross-floor recovery. Each task starts with an incorrect target-location prior in a region without the target. True target-category entries are removed from the runtime semantic map, while the physical scene and other map entries are preserved. Before evaluation, all initial candidate views are checked for target absence, and reachable views in another region are verified to detect a real target. Three fixed starting states and two repetitions yield 60 episodes per method, or 120 newly executed episodes. The ablated variant retains CPTG, JTVP, local multi-view reinspection, and the same perception and motion settings, but disables semantic-region recovery and its frontier fallback. Both groups use the same perturbed maps and starting states, a 240 s budget, and native caching.

Success requires exhausting the initial local search, leaving that region, and completing the task with RGB-D target evidence associated with a true same-category instance within 1 m. For SPL, the independently frozen A* reference route visits $\min(3, n_A)$ distinct initial viewpoints before a true-target candidate, where n_A is the initial candidate count. CT uses successful episodes' outer-process wall-clock times, including initialization, planning, and perception.

Table III shows that LogicFly succeeds in 36 of 60 episodes, whereas removing semantic recovery reduces SR to zero. All 60 full-method episodes trigger recovery after local search exhaustion; six tasks succeed in every starting state and repetition. The full method achieves an SPL of 42.43% and a successful-task CT of 12.78 s. Of its 24 unsuccessful episodes, 18 end without resolving the search and six report completion without passing the true-target geometric check. All 120 episodes finish without infrastructure failures or timeouts. These results demonstrate that semantic recovery enables cross-region search after local failure and substantially improves task completion when the initial target-location prior

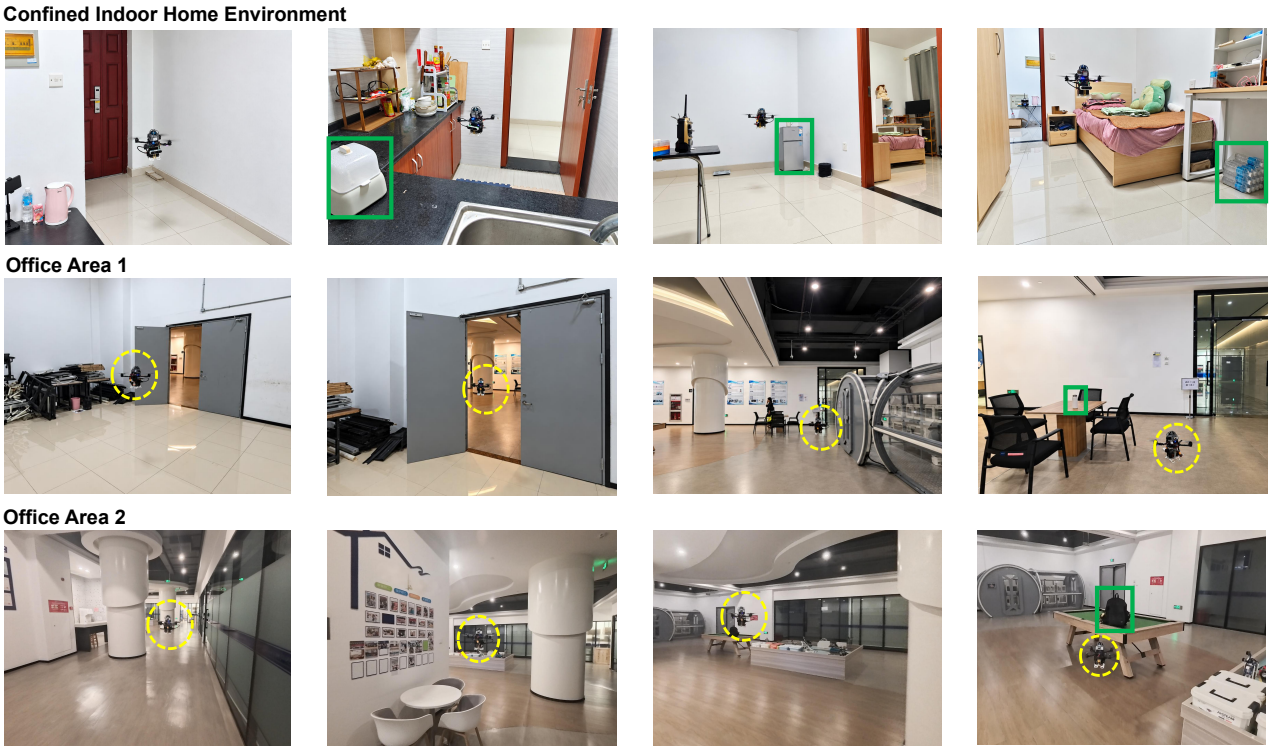

Fig. 9. Representative real-world UAV experiments in a confined indoor home environment (top row), Office Area 1 (middle row), and Office Area 2 (bottom row). The images show flight observations and visual detections during task execution.

is incorrect.

D. Evaluation in Real-World Environments

Experimental platform. We conduct real-world experiments using a custom-built quadrotor equipped with an NVIDIA Jetson Orin NX (16 GB), a PX4 flight controller, a Livox Mid-360 LiDAR, and an Intel RealSense D435 depth camera. The D435 provides synchronized 640×480 RGB and depth observations for online verification and 3D projection. FAST-LIO [43] provides onboard localization, while LiDAR-based occupancy mapping supports real-time trajectory optimization. The UAV is constrained to a maximum linear velocity of 1.0 m/s, linear acceleration of 1.0 m/s^2, and angular velocity of 1.0 rad/s.

Task completion and failures. We evaluate 10 long-horizon flight missions across five distinct real-world scenes. LogicFly successfully completes eight missions, yielding a task success rate of $8/10 = 80\%$. Both failed missions are attributed to errors in target recognition. These results demonstrate multi-task execution across different indoor environments and identify target recognition as the source of the observed mission failures.

Figure 1 illustrates a four-task mission: after failing to find the laptop at its instructed location, the UAV performs semantic recovery in the reading room, verifies the laptop, and completes the remaining checks. The map and numbered views document the executed route, recovery, and verification sequence.

Figure 9 shows representative flight observations and visual detections from the confined indoor home environment and the two office deployments.

VI. Conclusion

We presented **LogicFly**, a language-guided aerial navigation framework built around CPTG and JTVP. CPTG separates precedence and observation-triggered branches; JTVP selects task order and observation poses with 3D path and terminal costs within a bounded window. Verification and constrained recovery update subsequent decisions.

Across 854 episodes, LogicFly achieves 96.96% strict full-task SR and 0.8394 SPL, including 96.70% multi-target SR. Ablations verify CPTG's role in complex task logic. Compared with fixed task order and primary viewpoints, JTVP raises SPL from 76.82% to 82.15% and reduces successful-task CT from 12.62 s to 11.79 s at unchanged SR. Semantic recovery achieves 60.0% SR versus 0% without it on cross-region tasks, demonstrating its effectiveness under incorrect target-location priors. Deployment on a custom quadrotor further demonstrates multi-target execution with logical dependencies in real indoor rooms. These results advance VLN toward efficient closed-loop execution of logically complex missions.

References

[1] P. Anderson *et al.*, "Vision-and-language navigation: Interpreting visually-grounded navigation instructions in real environments," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2018, pp. 3674–3683.

[2] J. Krantz, E. Wijmans, A. Majumdar, D. Batra, and S. Lee, "Beyond the Nav-Graph: Vision-and-language navigation in continuous environments," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, ser. Lecture Notes Comput. Sci., vol. 12373, 2020, pp. 104–120.

[3] M. Shridhar *et al.*, "ALFRED: A benchmark for interpreting grounded instructions for everyday tasks," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 10 740–10 749.

[4] X. Song, W. Chen, Y. Liu, W. Chen, G. Li, and L. Lin, "Towards long-horizon vision-language navigation: Platform, benchmark and method," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 12 078–12 088.

[5] G. Zhou, Y. Hong, and Q. Wu, "NavGPT: Explicit reasoning in vision-and-language navigation with large language models," in *Proc. AAAI Conf. Artif. Intell.*, vol. 38, no. 7, 2024, pp. 7641–7649.

[6] J. Chen, B. Lin, R. Xu, Z. Chai, X. Liang, and K.-Y. K. Wong, "MapGPT: Map-guided prompting with adaptive path planning for vision-and-language navigation," in *Proc. 62nd Annu. Meeting Assoc. Comput. Linguistics (ACL), Vol. 1: Long Papers*, 2024, pp. 9796–9810.

[7] Z. Jiang and X. Wang, "SpatialGPT: Zero-shot vision-and-language navigation via spatial CoT over structured spatial memory," in *Proc. 33rd ACM Int. Conf. Adv. Geographic Inf. Syst. (SIGSPATIAL)*, 2025, pp. 423–435.

[8] C. Huang, O. Mees, A. Zeng, and W. Burgard, "Visual language maps for robot navigation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2023, pp. 10 608–10 615.

[9] Q. Gu *et al.*, "ConceptGraphs: Open-vocabulary 3D scene graphs for perception and planning," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 5021–5028.

[10] A. Werby, C. Huang, M. Büchner, A. Valada, and W. Burgard, "Hierarchical open-vocabulary 3D scene graphs for language-grounded robot navigation," in *Proc. Robot.: Sci. Syst. (RSS)*, 2024.

[11] A. Ku, P. Anderson, R. Patel, E. Ie, and J. Baldridge, "Room-Across-Room: Multilingual vision-and-language navigation with dense spatiotemporal grounding," in *Proc. Conf. Empir. Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 4392–4412.

[12] Y. Qi *et al.*, "REVERIE: Remote embodied visual referring expression in real indoor environments," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 9982–9991.

[13] F. Zhu, X. Liang, Y. Zhu, Q. Yu, X. Chang, and X. Liang, "SOON: Scenario oriented object navigation with graph-based exploration," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2021, pp. 12 689–12 699.

[14] W. Hao, C. Li, X. Li, L. Carin, and J. Gao, "Towards learning a generic agent for vision-and-language navigation via pre-training," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2020, pp. 13 137–13 146.

[15] Z. Wang *et al.*, "Scaling data generation in vision-and-language navigation," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2023, pp. 12 009–12 020.

[16] S. Chen, P.-L. Guhur, C. Schmid, and I. Laptev, "History aware multimodal transformer for vision-and-language navigation," in *Adv. Neural Inf. Process. Syst.*, vol. 34, 2021, pp. 5834–5847.

[17] S. Chen, P.-L. Guhur, M. Tapaswi, C. Schmid, and I. Laptev, "Think global, act local: Dual-scale graph transformer for vision-and-language navigation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 16 537–16 547.

[18] S. Wani, S. Patel, U. Jain, A. X. Chang, and M. Savva, "MultiON: Benchmarking semantic map memory using multi-object navigation," in *Adv. Neural Inf. Process. Syst.*, vol. 33, 2020, pp. 9700–9712.

[19] C. H. Song, J. Kil, T.-Y. Pan, B. M. Sadler, W.-L. Chao, and Y. Su, "One step at a time: Long-horizon vision-and-language navigation with milestones," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2022, pp. 15 482–15 491.

[20] J. Krantz *et al.*, "Iterative vision-and-language navigation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 14 921–14 930.

[21] M. Khanna *et al.*, "GOAT-Bench: A benchmark for multi-modal lifelong navigation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2024, pp. 16 373–16 383.

[22] F. L. Busch, T. Homberger, J. Ortega-Peimbert, Q. Yang, and O. Andersson, "One map to find them all: Real-time open-vocabulary mapping for zero-shot multi-object navigation," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 14 835–14 842.

[23] A. Rosinol, A. Gupta, M. Abate, J. Shi, and L. Carlone, "3D dynamic scene graphs: Actionable spatial perception with places, objects, and humans," in *Proc. Robot.: Sci. Syst. (RSS)*, 2020.

[24] N. Hughes, Y. Chang, and L. Carlone, "Hydra: A real-time spatial perception system for 3D scene graph construction and optimization," in *Proc. Robot.: Sci. Syst. (RSS)*, 2022.

[25] S. Peng, K. Genova, C. Jiang, A. Tagliasacchi, M. Pollefeys, and T. Funkhouser, "OpenScene: 3D scene understanding with open vocabularies," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2023, pp. 815–824.

[26] D. Shah, B. Osiński, B. Ichter, and S. Levine, "LM-Nav: Robotic navigation with large pre-trained models of language, vision, and action," in *Proc. 6th Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 205, 2023, pp. 492–504.

[27] B. Yu, H. Kasaei, and M. Cao, "L3MVN: Leveraging large language models for visual target navigation," in *Proc. IEEE/RSJ Int. Conf. Intell. Robots Syst. (IROS)*, 2023, pp. 3554–3560.

[28] J. Zhang, Z. Li, S. Wang, X. Shi, Z. Wei, and Q. Wu, "SpatialNav: Leveraging spatial scene graphs for zero-shot vision-and-language navigation," *arXiv preprint arXiv:2601.06806*, 2026.

[29] N. P. Bhatt *et al.*, "VLN-Zero: Rapid exploration and cache-enabled neurosymbolic vision-language planning for zero-shot transfer in robot navigation," *arXiv preprint arXiv:2509.18592*, 2025.

[30] Y. Qiao *et al.*, "Open-Nav: Exploring zero-shot vision-and-language navigation in continuous environment with open-source LLMs," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2025, pp. 6710–6717.

[31] X. Huang *et al.*, "MSGNav: Unleashing the power of multi-modal 3D scene graph for zero-shot embodied navigation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2026, pp. 37 154–37 163.

[32] K. Rana, J. Haviland, S. Garg, J. Abou-Chakra, I. Reid, and N. Sünderhauf, "SayPlan: Grounding large language models using 3D scene graphs for scalable robot task planning," in *Proc. 7th Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 229, 2023, pp. 23–72.

[33] Z. Dai *et al.*, "Optimal scene graph planning with large language model guidance," in *Proc. IEEE Int. Conf. Robot. Autom. (ICRA)*, 2024, pp. 14 062–14 069.

[34] A. Ray, C. Bradley, L. Carlone, and N. Roy, "Task and motion planning in hierarchical 3D scene graphs," *arXiv preprint arXiv:2403.08094*, 2024.

[35] W. Huang *et al.*, "Inner monologue: Embodied reasoning through planning with language models," in *Proc. 6th Conf. Robot Learn. (CoRL)*, ser. Proc. Mach. Learn. Res., vol. 205, 2023, pp. 1769–1782.

[36] N. Carion *et al.*, "SAM 3: Segment anything with concepts," *arXiv preprint arXiv:2511.16719*, 2025.

[37] Y. Zhang, X. Chen, C. Feng, B. Zhou, and S. Shen, "FALCON: Fast autonomous aerial exploration using coverage path guidance," *IEEE Trans. Robot.*, vol. 41, pp. 1365–1385, 2025.

[38] D. DeTone *et al.*, "Boxer: Robust lifting of open-world 2D bounding boxes to 3D," *arXiv preprint arXiv:2604.05212*, 2026.

[39] Y. Ren *et al.*, "Safety-assured high-speed navigation for MAVs," *Sci. Robot.*, vol. 10, no. 98, 2025, art. no. eado6187.

[40] Z. Wang, X. Zhou, C. Xu, and F. Gao, "Geometrically constrained trajectory optimization for multicopters," *IEEE Trans. Robot.*, vol. 38, no. 5, pp. 3259–3278, 2022.

[41] M. Savva *et al.*, "Habitat: A platform for embodied AI research," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2019, pp. 9339–9347.

[42] S. K. Ramakrishnan *et al.*, "Habitat-Matterport 3D Dataset (HM3D): 1000 large-scale 3D environments for embodied AI," in *Proc. Neural Inf. Process. Syst. Track Datasets Benchmarks*, vol. 1, 2021.

[43] W. Xu and F. Zhang, "FAST-LIO: A fast, robust LiDAR-inertial odometry package by tightly-coupled iterated Kalman filter," *IEEE Robot. Autom. Lett.*, vol. 6, no. 2, pp. 3317–3324, 2021.